Exploratory Data Analysis

This notebook presents an exploratory data analysis (EDA) of the MVTec Anomaly Detection MVTec AD[add source] dataset. MVTec AD is a benchmark dataset for unsupervised anomaly detection in industrial inspection and contains over 5,000 images across 15 object categories. Each category contains defect-free training images and a test set consisting of both normal and anomalous samples.

The purpose of this analysis is to:

1. Identify the object and texture categories present within the dataset.
2. Examine the distribution of normal and anomalous samples.
3. Investigate the defect types available within each category.
4. Analyse image characteristics such as resolution, brightness, sharpness, and contrast.
5. Identify potential challenges and limitations that may influence anomaly detection performance.
6. Inform preprocessing and model design decisions for the subsequent stages of the project.

1. Identify the object and texture categories present within the dataset.

bottle	cable	capsule	carpet	grid
hazelnut	leather	metal_nut	pill	screw
tile	toothbrush	transistor	wood	zipper

2. Examine the distribution of normal and anomalous samples.

This section examines how samples are distributed across the dataset. Understanding the balance between normal and anomalous images helps inform about any class imbalances may influence anomaly detection performance and evaluation metrics.

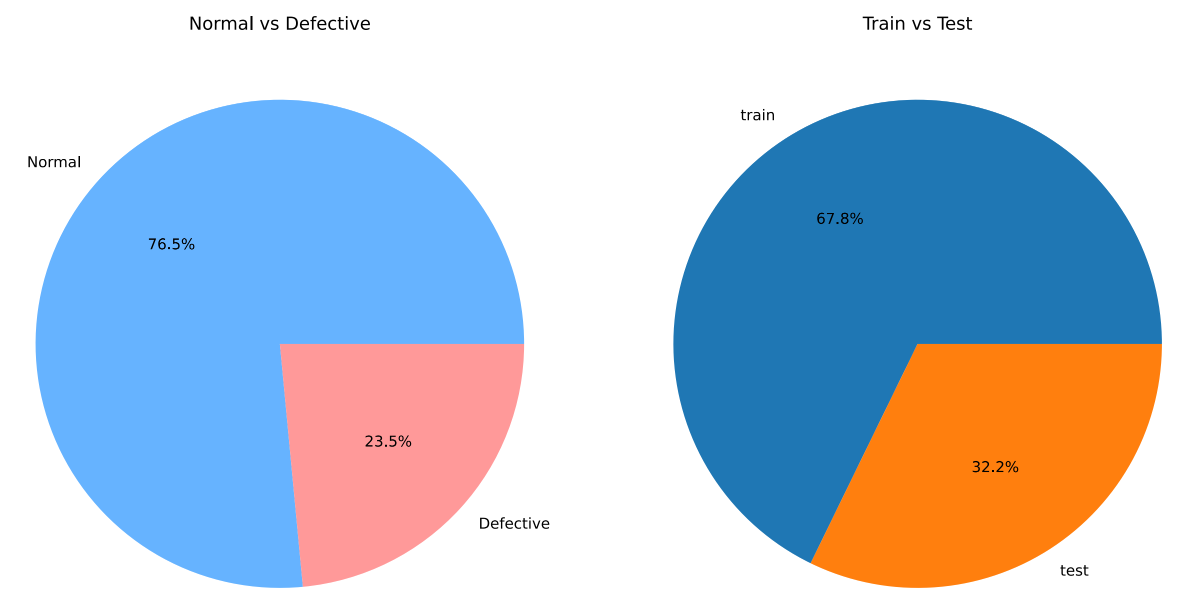

Figure 1:

2.1. Category Distribution

This investigates how images are distributed across the 15 MVTec categories. Identifying category-level imbalances helps determine whether certain categories may be overrepresented during evaluation.

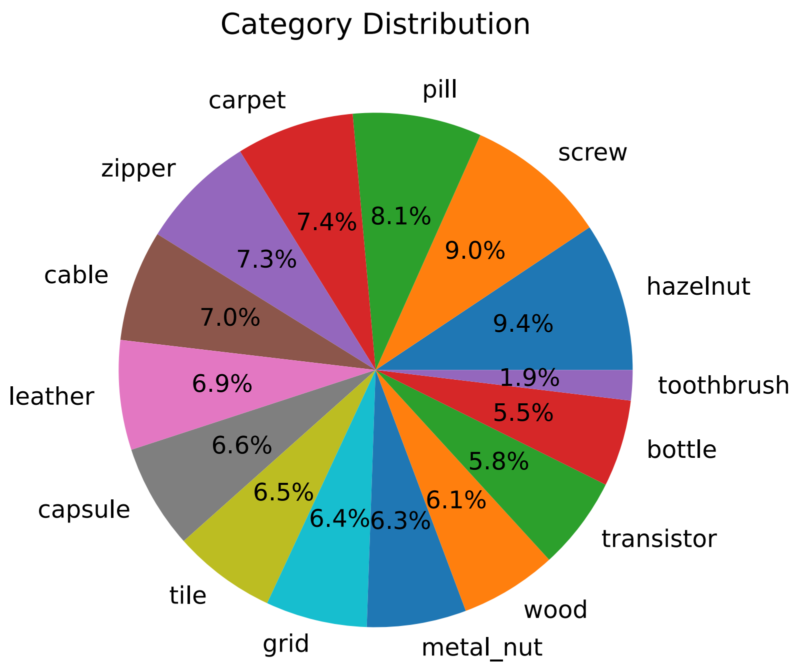

Figure 2:

The dataset is reasonably balanced across categories, although some categories contain more images than others. No category dominates the dataset to a degree that would significantly bias overall evaluation.

This figure shows the proportion of normal and defective samples within each category. As MVTec is designed for anomaly detection, categories may contain different ratios of normal and anomalous images depending on the number of defect types available.

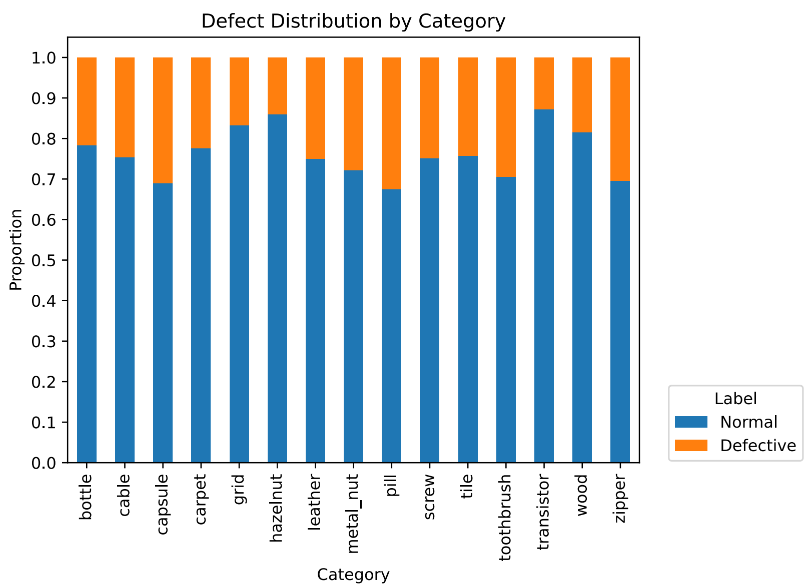

Figure 3:

The proportion of defective samples varies between categories. Categories containing a larger number of defect types generally show a higher proportion of anomalous test samples.

3. Investigate the defect types available within each category.

Understanding the variety of defect types present within each category is for assessing dataset complexity. Categories containing a larger number of defect types may present a more challenging anomaly detection task.

The following visualisations show how defect types are distributed within each category's test set. This provides insight into the relative frequency of different anomaly types.

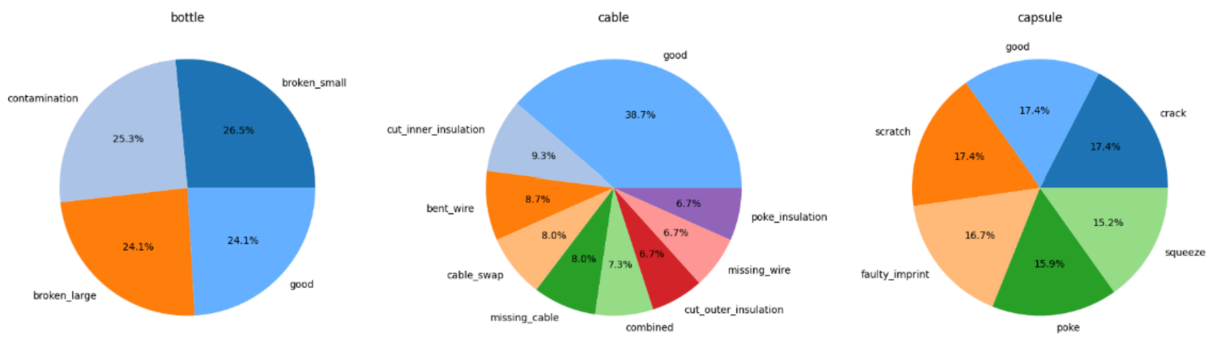

Figure 4: A sample of the distributions of defects in the data

The number and distribution of defect types varies substantially between categories. Some categories contain only a small number of anomaly classes, while others contain a wider range of defect variations.

This analysis examines the number of unique defect types present within each category. A larger number of defect types may indicate increased anomaly diversity and greater detection complexity.

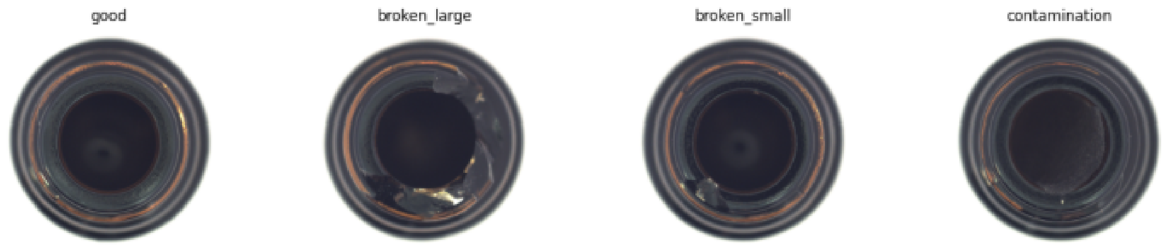

Figure 5: Example of defects in the bottle category

4. Analyse image characteristics such as resolution, brightness, sharpness, and contrast.

This section investigates several image characteristics that may influence feature extraction and anomaly detection performance. Resolution, brightness, sharpness, and contrast are analysed to identify potential inconsistencies or preprocessing requirements.

Image-level statistics are computed for every image in the dataset. Brightness is measured as the mean pixel intensity, contrast as the standard deviation of pixel intensities, and sharpness using the variance of the Laplacian.

Category	Brightness		Sharpness		Contrast	
	mean	std	mean	std	mean	std
bottle	136.74	1.71	44.38	4.63	92.72	0.95
cable	102.06	4.71	46.01	10.6	50.77	2.81
capsule	172.24	4.32	17.25	3.37	63.0	1.68
carpet	91.06	2.48	300.72	56.27	35.39	1.54
grid	114.11	9.23	30.09	11.4	39.18	7.52
hazelnut	51.72	4.13	43.01	3.84	26.49	5.13
leather	80.39	6.91	46.96	18.76	11.75	2.09
metal_nut	58.49	4.39	62.75	14.18	43.25	4.58
pill	76.33	4.96	142.1	96.86	78.45	4.05
screw	183.61	6.35	23.41	2.13	33.42	1.98
tile	116.35	8.45	46.61	14.97	31.5	2.23
toothbrush	47.02	8.02	167.17	137.77	55.81	1.61
transistor	79.52	3.44	63.58	8.43	42.07	1.56
wood	135.9	7.2	89.45	24.39	16.09	3.04
zipper	101.97	2.02	72.46	9.66	82.33	1.23

5. Observations

Several notable observations can be made from the image characteristic analysis:

- Image resolution is consistent within each category but varies between categories, ranging from 700×700 to 1024×1024 pixels.
- Brightness varies significantly between categories due to differences in object appearance and material properties, but remains highly consistent within individual categories.
- Sharpness and contrast exhibit greater variation between categories, reflecting differences in texture complexity and surface structure.
- Categories such as carpet, wood, and toothbrush contain substantially more texture detail than smoother categories such as capsule and screw.

These findings suggest that resizing and normalization will be necessary during preprocessing, while category-specific texture complexity may influence anomaly detection performance.